

- Q.19 Define cantilever. Write formula of time period of cantilever. (CO6)
- Q.20 Write any four properties of heat radiations. (CO5)
- Q.21 Check the correctness of (CO1)
- a) $v^2 - u^2 = 2as$ b) $F = ma$
- using dimensional analysis.
- Q.22 Define atmospheric pressure, absolute pressure and gauge pressure. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Define law of conservation of mechanical energy. Derive an expression for conservation of mechanical energy for freely falling bodies. (CO3)
- Q.24 Define Elasticity. Explain different types of modulus of elasticity. (CO4)
- Q.25 a) Explain centripetal force and centrifugal force with suitable example. (CO2)
- b) Derive an expression for calculating the total resistances when two resistors R_1 and R_2 are connected in series. (CO8)

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Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The dimensional formula of linear momentum is _____. (CO1)
- a) $[M^1 L^2 T^{-2}]$ b) $[M^3 L^2 T^{-3}]$
- c) $[M^3 L^{-1} T^{-2}]$ d) $[M^1 L^1 T^{-1}]$
- Q.2 The S.I. unit of work is _____. (CO3)
- a) Newton b) Volt
- c) Watt d) Joule
- Q.3 The value of 27°C of Kelvin scale is _____. (CO5)
- a) 300 K b) -273 K
- c) 273K d) 372K

Q.4 Frequency range of infrasonic waves is _____. (CO6)

- a) Less than 20Hz
- b) Between 20Hz to 20 kHz
- c) Greater than 20 kHz
- d) None of the above

Q.5 The substance through which current cannot flow are called _____. (CO9)

- a) Conductors b) Insulators
- c) Semiconductors d) None of these

Q.6 The emission responsible for producing laser light is _____. (CO10)

- a) Spontaneous emission
- b) Stimulated emission
- c) Thermal emission
- d) None of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define Gauss's law of electrostatics. (CO8)

Q.8 What is refraction of light? (CO7)

Q.9 Define simple harmonic motion. (CO6)

Q.10 Define convection of heat transfer. (Co6)

Q.11 State Hook's Law of elasticity. (CO4)

Q.12 Give formula of vector product of two vectors.(CO2)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 State Newton's second law of motion. Write equation of motion. (CO2)

Q.14 Derive formula used to convert one system of units into another system of units by using dimensional analysis. (CO1)

Q.15 Give four applications of optical fibers. (CO10)

Q.16 Define diamagnetic, paramagnetic and ferromagnetic materials. Give one example of each. (CO9)

Q.17 Write four differences between alternating current and direct current. (CO8)

Q.18 What is total internal reflection? Write essential conditions for total internal reflection. (CO7)